Topic 8: Order, Lattices, and Monotonicity

(How to do comparative statics without calculus)

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1. Why order shows up in economic theory

In economics, we care about how changes in the environment change outcomes. Because we aim to work on understanding human behavior, we care about how (equilibrium) choices change when the environment changes. The simplest way to say something here is what we call a comparative static, that is an answer to this question:

> "If an exogenous parameter goes up, does the optimal choice also go up?"

Examples:

- If income rises, does the consumer buy more of a good?
- If a bonus becomes more powerful, does effort increase?
- If one firm becomes more aggressive, do other firms respond by becoming more aggressive too?

Topic 7 separated two comparative-statics questions. The envelope theorem studies how the **optimized value** changes. Here we ask how the **optimizing choice** changes. As long as things are smooth enough, there is a straightforward way to deal with this second question (it still may not be an easy one, of course). We will briefly discuss that and then move on to the more general case which often is, unfortunately, also the more relevant one.

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2. Excursion: The Implicit Function Theorem

Suppose we live in a perfectly smooth world. Everything is as differentiable and concave as we want it to be. Moreover, for the sake of this argument, let's assume we are working on a single dimension. So, what we are looking at is a function $f(x;\theta)$ where x is our variable of choice, and θ is a parameter. Now, we know the necessary and sufficient condition for an optimum is that

$$f'(x;\theta) := \frac{\partial f(x;\theta)}{\partial x} = 0.$$

Solving this for x gives us our optimum x^*. Now if we change θ, that optimum is likely to change, but[1] we know that even under another θ' it must be true, that x'^* solves

[1] still assuming the first-order conditions are necessary and sufficient in our problem

$$f'(x'^*;\theta') = 0.$$

Knowing this, means, we know also how the first order condition needs to change if we change θ slightly, simply by using the fact that although x is our choice and θ is a parameter (if you like the *state* of the problem), mathematically f is a function of x and θ. Moreover, we can (in principle at least) describe $x(\theta)$ as the choice function that describes how the optimal choice changes in θ.

So in our example problem $x(\theta) = x^*$ and $x(\theta') = x'^*$. Hopefully, you now see where this is going. The object we want to understand fundamentally is $\partial x(\theta)/\partial\theta$ or in words, how does our choice x change if the environment changes slightly.[2]

One complication could be that if f is complicated, explicitly describing $x(\theta)$ in *closed form*[3] may be difficult.[4]

But, our knowledge, that x (whatever it is) has to solve the first-order condition tells us *implicitly* what x is, and more relevantly, it tells us *how it changes in* θ. Why the latter?

Well, think about the function $f(x(\theta);\theta)$. For that function, we know that

$$f'(x(\theta);\theta) = 0$$

So if $\hat{x} := x(\hat{\theta})$ for some $\hat{\theta}$, then we have $f'(\hat{x},\hat{\theta}) = 0$ and as long as we keep it this way, this is always true. Then, for small changes $\mathrm{d}\theta$, that means[5]

$$\mathrm{d}f'(x;\theta) = \left(\frac{\partial f'(x;\theta)}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f'(x;\theta)}{\partial \theta}\right)\mathrm{d}\theta = 0$$

if we want to ensure that we stay on the "solution" part in x.[6]

But this can only be true if the part in the large brackets is 0, which in turn means that

$$\frac{\partial x}{\partial \theta} = -\frac{\frac{\partial f'(x;\theta)}{\partial \theta}}{\frac{\partial f'(x;\theta)}{\partial x}}.$$

[2]For example, if unemployment benefit (θ) goes up, how does the labor supply respond (does it go up? How much does it go up?).

[3]That means as an actual function.

[4]For example, think of a first order condition $xe^{\theta x} - 3 = 0$. There is no closed-form in x.

[5]Important: Note we are taking derivatives of the *derivative* here! Not of the original function!

[6]We implicitly assume here that $\partial f'(x;\theta)/\partial x \neq 0$ (otherwise we divide by zero). But recall, that for a (strictly) concave function that is no problem.

This last relation is called the *implicit function theorem* as it tells us that just knowing the first order equation, is enough to understand how the optimal choice, x, changes in the parameter θ.

In the example where the first order condition is

$$xe^{\theta x} - 3 = 0$$

we know there is no closed form solution $x(\theta)$,[7] yet we can solve

$$\frac{\partial x(\theta)}{\partial \theta} = -\frac{(x(\theta))^2}{1 + \theta x(\theta)}$$

via the implicit function theorem.

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3. Back to general problems

The above works nicely under great simplifying assumptions, but they can fail for reasons we have already seen:

- the choice may be multidimensional,
- the solution may be a set, not a point,
- the objective may not be smooth (kinks are common),
- the action space may be discrete ("choose a technology" is not a calculus problem).

The order-theoretic approach we will now discuss is the alternative. It is designed for exactly the situations where you do not want (or cannot) to lean on calculus.

The punchline of this topic is:

> If you can express your model using an order ("higher" versus "lower") and you can show a certain kind of complementarity, then monotonicity results fall out cleanly.

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4. Partially ordered sets

Let X be a set. A binary relation $\preceq$ on X is a subset of $X \times X$ (Topic 2 already used this viewpoint for preferences).

We say that $(X, \preceq)$ is a **partially ordered set** (a "poset") if $\preceq$ satisfies:

[7] You can solve it with the quasi-closed-form usage of the Lambert-W. We will not get into those details.

1. **Reflexivity:** for every $x \in X$,

$$x \preceq x.$$

Every element is weakly below itself.

2. **Antisymmetry:** for all $x, y \in X$,

$$x \preceq y \text{ and } y \preceq x \;\Rightarrow\; x = y.$$

If each is weakly below the other, then they are the same element.

3. **Transitivity:** for all $x, y, z \in X$,

$$x \preceq y \text{ and } y \preceq z \;\Rightarrow\; x \preceq z.$$

If x is below y and y is below z, then x is below z.

A partial order is "partial" because not every pair can be compared under $\preceq$.

Total orders

A poset is a **total order** (sometimes called "linear order") if any two elements can be compared: for all $x, y \in X$,

$$x \preceq y \quad \text{or} \quad y \preceq x.$$

The real line $(\mathbb{R}, \leq)$ is a total order.

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5. Two orders economists use all the time

5.1 The usual order on $\mathbb{R}$

This is the one you already know: $x \leq y$.

5.2 The product order on $\mathbb{R}^n$

In consumer problems, choices are usually vectors. So we need an order for vectors.

For $x, y \in \mathbb{R}^n$, define

$$x \preceq y \quad \Longleftrightarrow \quad x_i \leq y_i \text{ for all } i = 1, \ldots, n.$$

"y is higher than x if it is at least as high in every coordinate.

This is a partial order, not a total order. For instance, in $\mathbb{R}^2$, $(1, 0)$ and $(0, 1)$ are not comparable under $\preceq$.

Economically, the product order is the natural order when "more of every component" is interpreted as "bigger".

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6. Monotone maps

Let $(X, \preceq_X)$ and $(Y, \preceq_Y)$ be posets.

A function $g : X \to Y$ is **monotone increasing** if

$$x \preceq_X x' \;\Rightarrow\; g(x) \preceq_Y g(x').$$

"If you increase the input, the output does not go down."

A function g is **monotone decreasing** if

$$x \preceq_X x' \;\Rightarrow\; g(x') \preceq_Y g(x).$$

These definitions are deliberately minimal. They do not require differentiability, continuity, or convexity. They only require that the words "higher" and "lower" make sense.

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7. Lattices: when any two points have a meet and a join

A poset $(X, \preceq)$ is a **lattice** if for any two elements $x, y \in X$ there exist:

- a **join** (least upper bound), written $x \vee y$, and
- a **meet** (greatest lower bound), written $x \wedge y$,

such that:

1. $x \preceq x \vee y$ and $y \preceq x \vee y$ (it is an upper bound), and if z is any upper bound of both x and y, then $x \vee y \preceq z$ (it is the least upper bound);
2. $x \wedge y \preceq x$ and $x \wedge y \preceq y$ (it is a lower bound), and if z is any lower bound of both x and y, then $z \preceq x \wedge y$ (it is the greatest lower bound).

In English:

- $x \vee y$ (the least upper bound) is "the smallest element that is above both x and y,"
- $x \wedge y$ (the greatest lower bound) is "the largest element that is below both x and y."

What does it mean, practically? It means in particular, that any element z in X that is (under our order) larger than both x and y, then it has to be ranked with (and larger than) a single, defined *join*. And analogously for the meet. Important, this does *not* imply a total order, as we will see in this (often relevant) example.

Example: $\mathbb{R}^n$ with the product order

Under the product order on $\mathbb{R}^n$,

$$(x \vee y)_i = \max\{x_i, y_i\}, \qquad (x \wedge y)_i = \min\{x_i, y_i\}.$$

So the n–dimensional Euclidean space under the product order is indeed a lattice.

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8. Complete lattices (and why you might care later)

A lattice $(X, \preceq)$ is a **complete lattice** if every subset $A \subseteq X$ has:

- a least upper bound, and
- a greatest lower bound.

"A lattice is complete if we not only can take joins and meets of two points; but we can take them for any collection of points."

Why this matters (preview): complete lattices are the right environment for fixed-point results like **Tarski's fixed-point theorem**, which is a workhorse in monotone equilibrium arguments.

In terms of notation, we call the least upper bound of a set its *supremum* and the greatest lower bound its *infimum*. **Important:** As in the two-point case, there is nothing that says these bounds need to lie *in the subset*, it just says that any point that we can compare to the elements of the set, we can also compare to those, and the order is in the right way.

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9. Increasing differences: the core monotonicity condition

Now we move from ordering choices to ordering how incentives change.

Let X be a poset (often a lattice) and let T be a poset of parameters (often $T = \mathbb{R}$ with $\leq$).

Consider an objective function

$$f : X \times T \to \mathbb{R}.$$

We say that f has **increasing differences** in (x, t) if for all $x' \succeq x$ and $t' \succeq t$,

$$f(x', t') - f(x, t') \geq f(x', t) - f(x, t).$$

In English:

> "f has increasing differences if, when the parameter t is higher, increasing x becomes (weakly) more attractive."

There is a very concrete way to think about it:

- fix two actions $x \preceq x'$,
- compare the advantage of choosing x' rather than x,
- increasing differences says that this advantage does not fall when t rises.

A quick differentiable sanity check (only as intuition)

If X and T are intervals in $\mathbb{R}$ and f is twice differentiable, increasing differences corresponds to a nonnegative cross-partial:

$$\frac{\partial^2 f}{\partial x\,\partial t}(x,t) \geq 0.$$

This is not the definition, but it is a useful mental picture: cross-partials capture complementarity.

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10. Supermodularity: complementarity inside the choice vector

Increasing differences is about complementarity between a choice x and a parameter t.

Supermodularity is about complementarity between components of the choice vector itself.

Let X be a lattice and $f : X \to \mathbb{R}$.

We say that f is **supermodular** if for all $x, y \in X$,

$$f(x \vee y) + f(x \wedge y) \geq f(x) + f(y).$$

In English:

"If I take the coordinatewise higher bundle and the coordinatewise lower bundle, their total value is at least as large as the total value of the original two bundles."

This inequality encodes "components like to move together." In $\mathbb{R}^n$, it is closely related to nonnegative cross-effects (or cross-partial derivatives).

(Again: cross-partials are the differentiable picture, not the definition.)

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11. Topkis' theorem: monotone comparative statics without derivatives

We now connect this back to Topic 5, where we treated optimal choices as correspondences.

Let X be a lattice, T a poset, and let $f : X \times T \to \mathbb{R}$.

For each $t \in T$, define the argmax correspondence

$$X^*(t) = \text{argmax}_{x \in X}\, f(x,t).$$

Topkis' monotonicity theorem (one standard version) says:

If (i) $f(\cdot, t)$ is supermodular on X for each t, and (ii) f has increasing differences in (x, t), then the set of maximizers moves upward with t in the following sense:

- If $t' \succeq t$, and $x \in X^*(t)$, $x' \in X^*(t')$, then there exist maximizers $\underline{x} \in X^*(t)$ and $\overline{x} \in X^*(t')$ such that

$$\underline{x} \preceq \overline{x}.$$

In English:

> "Higher parameters do not force optimal choices to move down. Even if the optimizer is not unique, you can choose maximizers consistently so that they move upward."

This is why the lattice approach is so useful: you can get monotone comparative statics even when:

- the solution is set-valued,
- the objective is not differentiable,
- you do not want to solve the model explicitly.

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12. A single-agent economic example: effort choice with a bonus parameter

An agent chooses effort e from an interval $X = [0, 1]$ (ordered by $\leq$). A principal chooses a bonus level $b \in T = \mathbb{R}_+$.

Suppose the agent's payoff is

$$f(e, b) = b \cdot e - c(e),$$

where c is increasing and convex.

Interpretation:

- $b \cdot e$ is the marginal reward from effort,
- $c(e)$ is the cost.

Now check increasing differences.

Take $e' > e$ and $b' > b$. Then

$$f(e', b') - f(e, b') = b'(e' - e), \qquad f(e', b) - f(e, b) = b(e' - e).$$

Since $b' > b$ and $e' - e > 0$, we have

$$b'(e' - e) \geq b(e' - e),$$

so f has increasing differences in (e, b).

What it means:

> "A higher bonus makes higher effort relatively more attractive."

So the optimal effort correspondence $E^*(b) = \operatorname{argmax}_{e \in [0,1]} f(e, b)$ is (weakly) increasing in b.

Notice what we did not use:

- no first-order conditions,
- no envelope theorem,
- no differentiation of a solution function.

We only used the order structure and a simple algebraic inequality.

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13. Outlook: why this matters for micro (and for games later)

Order and lattice methods matter because they let you prove robust qualitative statements that survive model variations.

In microeconomics, they show up in at least three places:

1. **Monotone comparative statics in optimization**
   This is what we did: higher parameters lead to higher optimal choices under complementarity.

2. **Strategic complements in games**
   When players' best responses are increasing, equilibrium sets often have lattice structure. This is one route to existence and to sharp predictions about how equilibrium moves with parameters.

3. **Mechanism design and auctions**
   "Single-crossing" and monotonicity of optimal reports are close cousins of increasing differences. They explain why allocation rules tend to be monotone in types.

So this topic is not just a mathematical curiosity. It is the vocabulary behind a large family of economically meaningful monotonicity results.

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14. Where this appears in MWG (and where it does not)

- The language of relations and orders is closest in spirit to **MWG, Mathematical Appendix A** (relations and basic set-theoretic language).
- The lattice-theoretic approach to monotone comparative statics (increasing differences, supermodularity, Topkis' theorem) is not developed systematically in MWG's mathematical appendices.

For this topic, MWG is best treated as background micro motivation, while lattice methods are typically learned from dedicated comparative statics or game-theory notes.

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15. What you should take away

After this topic, you should be comfortable with:

- reading statements like $x \preceq x'$ and understanding what order is being used,
- recognizing $\mathbb{R}^n$ with the product order as the natural "more in every dimension" comparison,
- understanding increasing differences as "higher parameters strengthen incentives to choose higher actions,"
- seeing why monotone comparative statics is compatible with set-valued argmax correspondences.

Once this is in place, supermodular games and monotone equilibrium arguments will feel like a continuation rather than a new language.